Ensemble-Based Spatial Snow Water Equivalent Estimates Using the SWE Evaluation Tool for the Rio Grande Basin

March 08, 2026

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Introduction

No spaceborne sensor can directly measure snow water equivalent (SWE). Therefore, the spatial distribution of SWE must be estimated from remotely sensed observations combined with modeling approaches. Existing gridded SWE products exhibit documented accuracy limitations across both mountainous and non-mountainous regions, resulting in a wide range of SWE estimates for any given location. To address this uncertainty, the New Mexico Office of the State Engineer (NM OSE), in collaboration with the Institute of Arctic and Alpine Research (INSTAAR), developed the SWE Evaluation Tool (SWEET). SWEET integrates multiple suitable SWE data products into an ensemble framework, with the objective of producing near-real-time spatial SWE estimates for the Rio Grande Basin that are, on average, as accurate as or more accurate than any individual product alone.

Figure 1 shows the near-real-time (March 08, 2026) ensemble SWE estimates for the Rio Grande Basin. Subsequent sections of this report present detailed ensemble-based SWE maps, along with summaries by subbasin and elevation band. In addition to the ensemble results highlighted in this report, figures and tables for each individual SWE product are also provided for download to support transparency and further analysis. These data sets and visualizations are available on the GitHub repository.

For ease of navigation, hyperlinks are provided throughout this report to help readers quickly access basin-specific sections and supporting data tables.

Rio Grande Basin SWE Overview

Rio Grande Headwaters

Upper Rio Grande

Rio Grande-Elephant Butte (Middle Rio Grande)

About SWEET

Rio Grande Basin SWE Overview

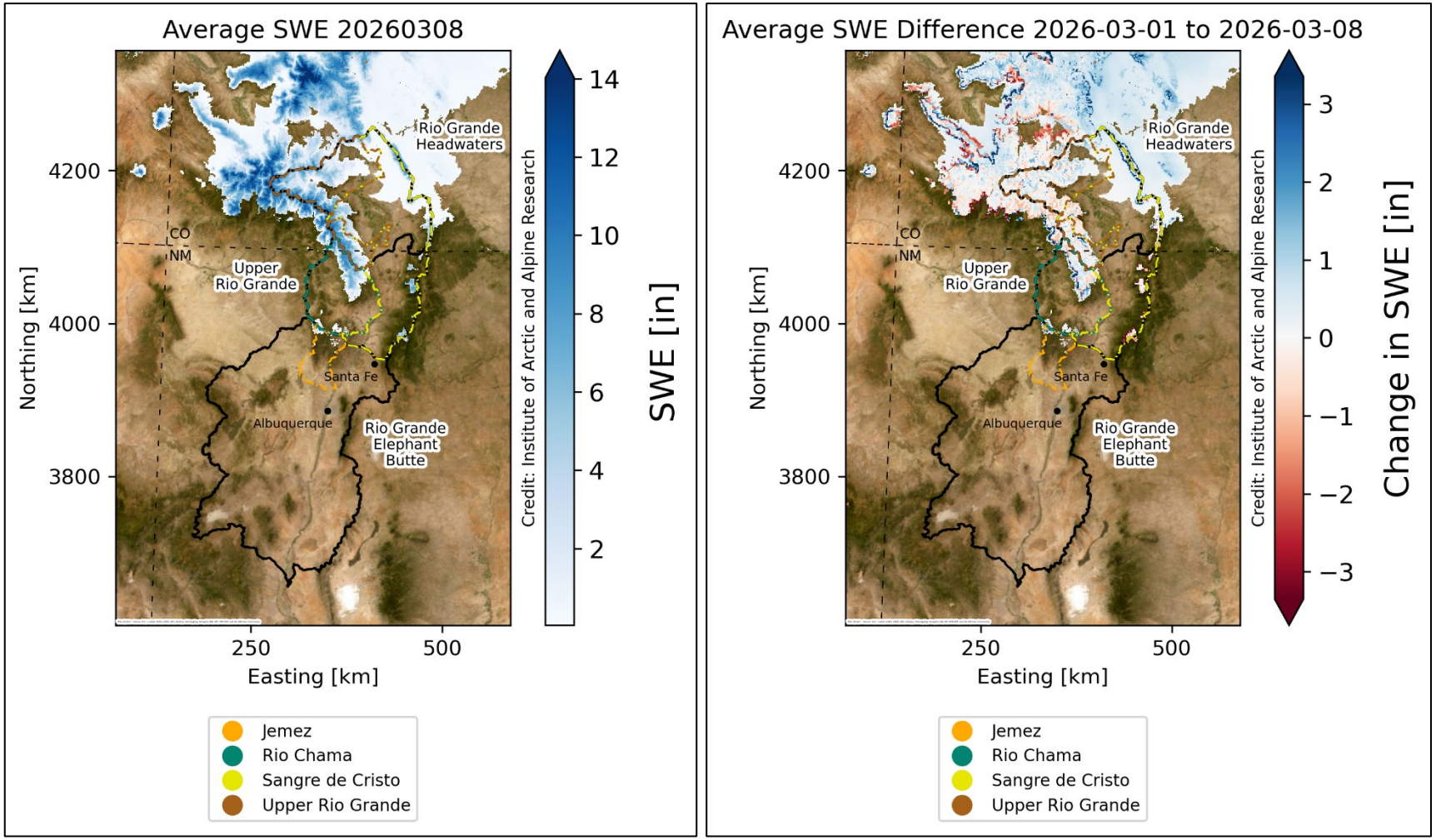

Figure 1. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande Basin for March 08, 2026. *SWE amounts (in.) across the Rio Grande Basin (left); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande Basin from March 01, 2026, to March 08, 2026 (right); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts.*

Table 1. Estimated ensemble SWE by HUC6 Basin for March 08, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since March 01, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value. Values are separated by HUC8 subbasin and totaled for each HUC8 basin (bold).*

HUC6 Basin	HUC8 Basin	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-01 (af)	Median Vol. Change since 2026-03-01 (af)
Rio Grande Headwaters	Rio Grande Headwaters	338,857	334,872	313,069 (UA_SWE)	368,629 (CU_SWE)	1,113	-1,749
	Alamosa-Trinchera	103,473	107,841	86,500 (SNODAS)	116,079 (CU_SWE)	-3,181	-4,285
	San Luis	68,687	67,112	64,045 (CU_SWE)	74,903 (UA_SWE)	12,697	11,623
	Saguache	69,290	65,271	62,268 (SNODAS)	80,331 (UA_SWE)	-1,432	-3,123
	Conejos	145,781	144,463	131,894 (UA_SWE)	160,986 (SNODAS)	427	7,662
	Total	**726,088**	**711,738**	**708,038 (UA_SWE)**	**758,487 (CU_SWE)**	**9,624**	**-315**
Upper Rio Grande	Upper Rio Grande	72,624	74,356	47,616 (CU_SWE)	95,899 (SNODAS)	-12,591	-16,122
	Rio Chama	144,599	140,330	105,146 (UA_SWE)	188,322 (SNODAS)	-3,472	5,877
	Total	**217,223**	**187,945**	**179,502 (UA_SWE)**	**284,220 (SNODAS)**	**-16,063**	**-16,250**
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	741	389	361 (CU_SWE)	1,472 (SNODAS)	-99	-112
	Jemez	4,365	4,341	3,269 (UA_SWE)	5,486 (CU_SWE)	-1,373	-2,510
	Rio Grande-Albuquerque	8	0	0 (SNODAS, UA_SWE)	24 (CU_SWE)	2	0
	Arroyo Chico	0	0	0	0	-22	0
	North Plains	0	0	0	0	0	0
	Rio San Jose	0	0	0	0	-118	0
	Plains of San Agustin	0	0	0	0	0	0
	Rio Salado	0	0	0	0	0	0
	Jornada Del Muerto	0	0	0	0	0	0
	Elephant Butte Reservoir	0	0	0	0	-18	0
	Rio Puerco	1,355	1,082	1,051 (SNODAS)	1,931 (CU_SWE)	-435	-819
	Total	**6,469**	**6,864**	**4,740 (UA_SWE)**	**7,802 (CU_SWE)**	**-2,063**	**-3,348**

Table 2. Estimated ensemble SWE by specified subbasin groupings for March 08, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since March 01, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value. Values are separated by HUC8 subbasin and totaled for each specified subbasin grouping (bold).*

OSE Grouping	HUC8 Basin	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-01 (af)	Median Vol. Change since 2026-03-01 (af)
Upper Rio Grande	Rio Grande Headwaters	338,857	334,872	313,069 (UA_SWE)	368,629 (CU_SWE)	1,113	-1,749
	Saguache	69,290	65,271	62,268 (SNODAS)	80,331 (UA_SWE)	-1,432	-3,123
	Conejos	145,781	144,463	131,894 (UA_SWE)	160,986 (SNODAS)	427	7,662
	Total	**553,928**	**558,126**	**525,294 (UA_SWE)**	**578,363 (CU_SWE)**	**108**	**1,621**
Sangre de Cristo	San Luis	68,687	67,112	64,045 (CU_SWE)	74,903 (UA_SWE)	12,697	11,623
	Alamosa-Trinchera	103,473	107,841	86,500 (SNODAS)	116,079 (CU_SWE)	-3,181	-4,285
	Upper Rio Grande	72,624	74,356	47,616 (CU_SWE)	95,899 (SNODAS)	-12,591	-16,122
	Total	**244,784**	**249,511**	**227,740 (CU_SWE)**	**257,100 (UA_SWE)**	**-3,075**	**-14,323**
Rio Chama	**Total**	**144,599**	**140,330**	**105,146 (UA_SWE)**	**188,322 (SNODAS)**	**-3,472**	**5,877**
Jemez	**Total**	**4,365**	**4,341**	**3,269 (UA_SWE)**	**5,486 (CU_SWE)**	**-1,373**	**-2,510**

Upper Rio Grande Snowpack Update Application

The OSE/ISC Upper Rio Grande Snowpack Update Application (Application) provides water managers and users with detailed SNOTEL-based SWE data organized by the same subbasin groupings shown in Table 2. The Application is updated regularly from December through May and compiles key SWE information from SNOTEL sites within each subbasin. In addition to the Upper Rio Grande groupings (Upper Rio Grande, Sangre de Cristo, Rio Chama, and Jemez), the Application also includes the San Juan River Basin, with data summarized for the Upper San Juan, Animas, and Piedra subbasins.

The Application can be viewed here[1].

[1] Please note that access to the Application may be restricted to employees of the NM OSE/ISC at this time. We are currently working on public access.

Rio Grande Headwaters

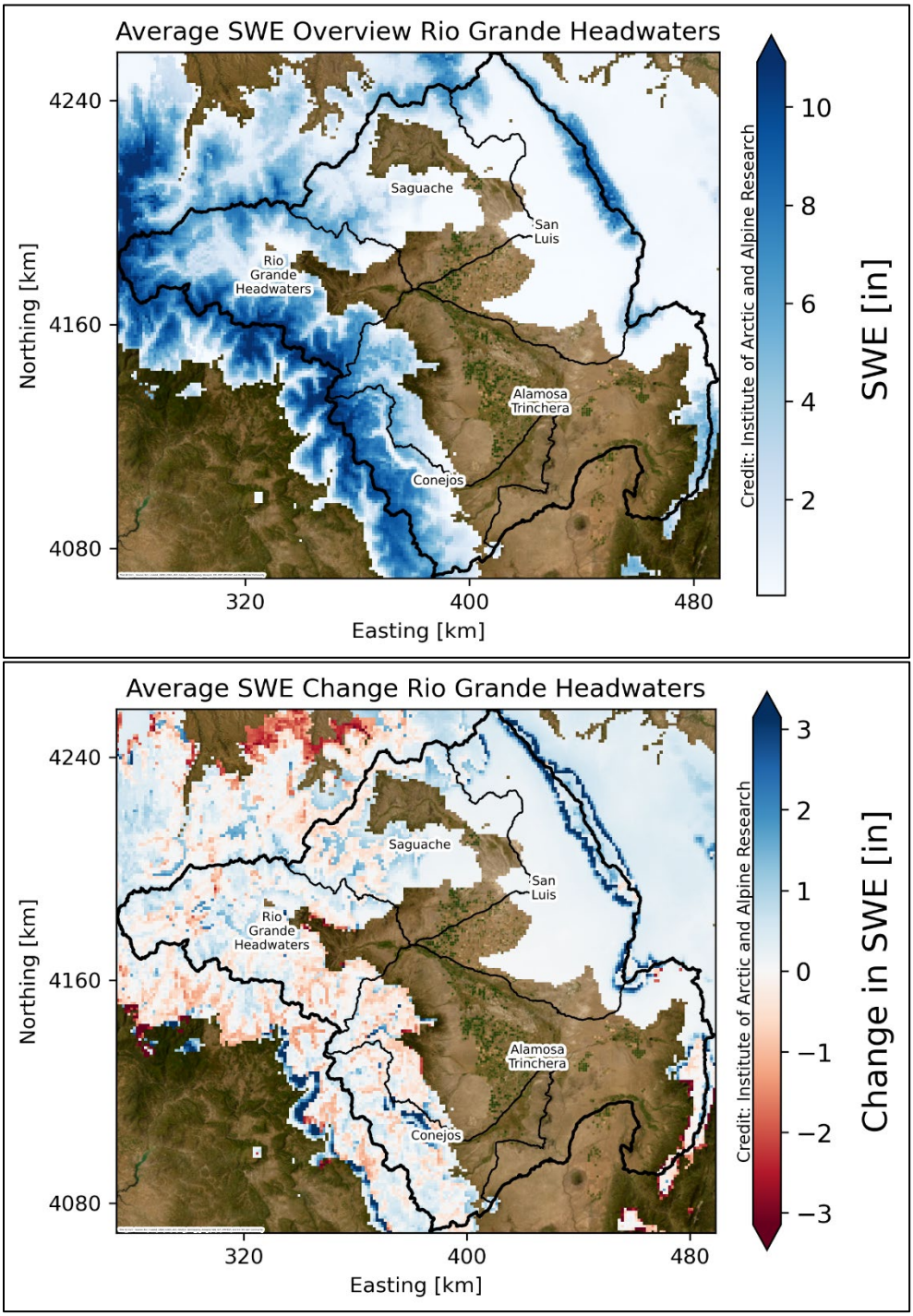

Figure 2. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande Headwaters HUC6 Basin for March 08, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Rio Grande Headwaters Basin, showing spatial patterns of SWE across subbasins including Rio Grande Headwaters, Alamosa-Trinchera, San Luis, Saguache, and Conejos (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande Headwaters from March 01, 2026, to March 08, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

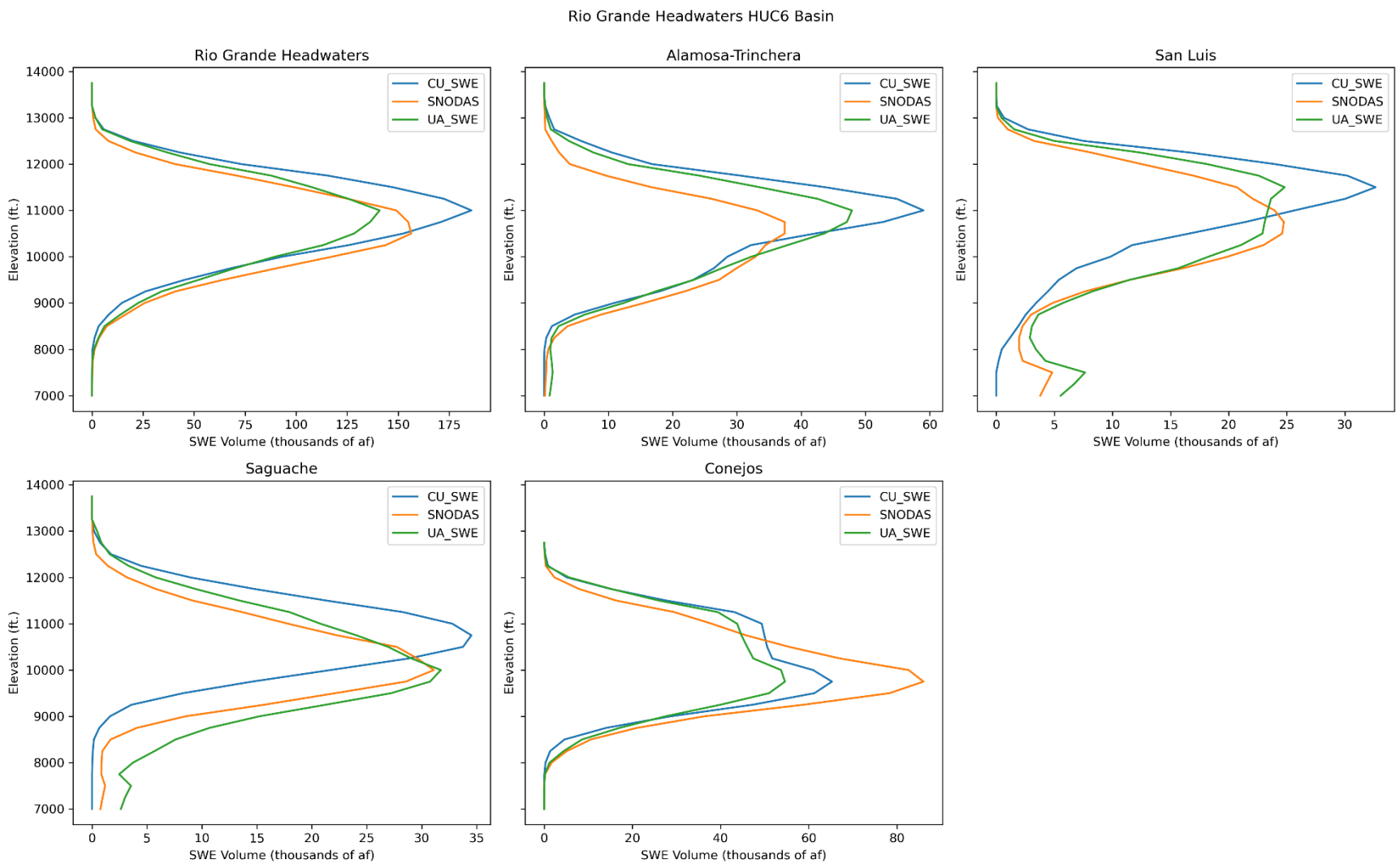

Figure 3. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Rio Grande Headwaters HUC6 Basin for March 08, 2026. *Elevation-distributed SWE volume for the Rio Grande Headwaters HUC6 Basin and associated subbasins (Rio Grande Headwaters, Alamosa–Trinchera, San Luis, Saguache, and Conejos), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 3. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Rio Grande Headwaters HUC6 Basin for March 08, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since March 01, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-01 (af)	Median Vol. Change since 2026-03-01 (af)
Rio Grande Headwaters	Rio Grande Headwaters	7000-8000'	0	0	0	0	0	0
		8000-9000'	781	889	203 (CU_SWE)	1,251 (SNODAS)	-693	-933
		9000-10000'	20,954	22,576	14,540 (CU_SWE)	25,746 (SNODAS)	-3,074	-2,067
		10000-11000'	100,129	93,219	89,581 (UA_SWE)	117,588 (SNODAS)	-3,594	-5,273
		11000-12000'	158,626	149,088	140,916 (UA_SWE)	185,874 (CU_SWE)	4,541	196
		12000-13000'	57,079	57,473	40,632 (SNODAS)	73,132 (CU_SWE)	3,810	882
		13000-14000'	1,287	1,633	567 (SNODAS)	1,661 (CU_SWE)	123	309
	Alamosa-Trinchera	7000-8000'	327	128	0 (CU_SWE)	852 (UA_SWE)	327	128
		8000-9000'	546	642	50 (CU_SWE)	947 (UA_SWE)	-247	576
		9000-10000'	12,933	12,366	10,892 (CU_SWE)	15,541 (SNODAS)	-5,606	-4,773
		10000-11000'	31,229	32,252	28,517 (CU_SWE)	32,916 (SNODAS)	-2,690	-3,331
		11000-12000'	46,698	47,901	33,161 (SNODAS)	59,033 (CU_SWE)	3,130	607
		12000-13000'	11,274	13,054	3,990 (SNODAS)	16,778 (CU_SWE)	1,801	1,658
		13000-14000'	466	469	121 (SNODAS)	809 (CU_SWE)	105	12
	San Luis	7000-8000'	3,112	3,790	0 (CU_SWE)	5,546 (UA_SWE)	3,087	3,790
		8000-9000'	1,953	1,971	476 (CU_SWE)	3,412 (UA_SWE)	1,456	1,724
		9000-10000'	4,668	4,847	3,429 (CU_SWE)	5,729 (UA_SWE)	705	1,220
		10000-11000'	16,000	18,259	9,835 (CU_SWE)	19,907 (SNODAS)	1,228	896
		11000-12000'	24,332	23,951	23,369 (UA_SWE)	25,677 (CU_SWE)	2,992	2,373
		12000-13000'	18,205	18,142	12,511 (SNODAS)	23,963 (CU_SWE)	3,132	1,334
		13000-14000'	415	446	136 (SNODAS)	664 (CU_SWE)	97	29
	Saguache	7000-8000'	1,129	760	0 (CU_SWE)	2,626 (UA_SWE)	1,126	760
		8000-9000'	1,533	859	21 (CU_SWE)	3,721 (UA_SWE)	-231	803
		9000-10000'	8,448	8,495	1,631 (CU_SWE)	15,217 (UA_SWE)	-1,036	4,083
		10000-11000'	28,184	31,102	21,709 (CU_SWE)	31,742 (UA_SWE)	-3,287	-1,089
		11000-12000'	23,802	20,780	17,835 (SNODAS)	32,790 (CU_SWE)	1,272	108
		12000-13000'	5,970	5,787	3,180 (SNODAS)	8,943 (CU_SWE)	703	172
		13000-14000'	224	176	37 (SNODAS)	458 (UA_SWE)	20	45
	Conejos	7000-8000'	0	0	0	0	0	0

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-01 (af)	Median Vol. Change since 2026-03-01 (af)
Rio Grande Headwaters	Conejos	8000-9000'	1,076	1,221	309 (CU_SWE)	1,698 (SNODAS)	-1,028	-725
		9000-10000'	30,806	28,655	27,489 (UA_SWE)	36,275 (SNODAS)	-2,314	-2,878
		10000-11000'	65,802	60,991	53,741 (UA_SWE)	82,675 (SNODAS)	1,483	5,780
		11000-12000'	43,719	43,783	38,008 (SNODAS)	49,365 (CU_SWE)	1,978	808
		12000-13000'	4,378	5,142	2,331 (SNODAS)	5,660 (UA_SWE)	308	892

Upper Rio Grande

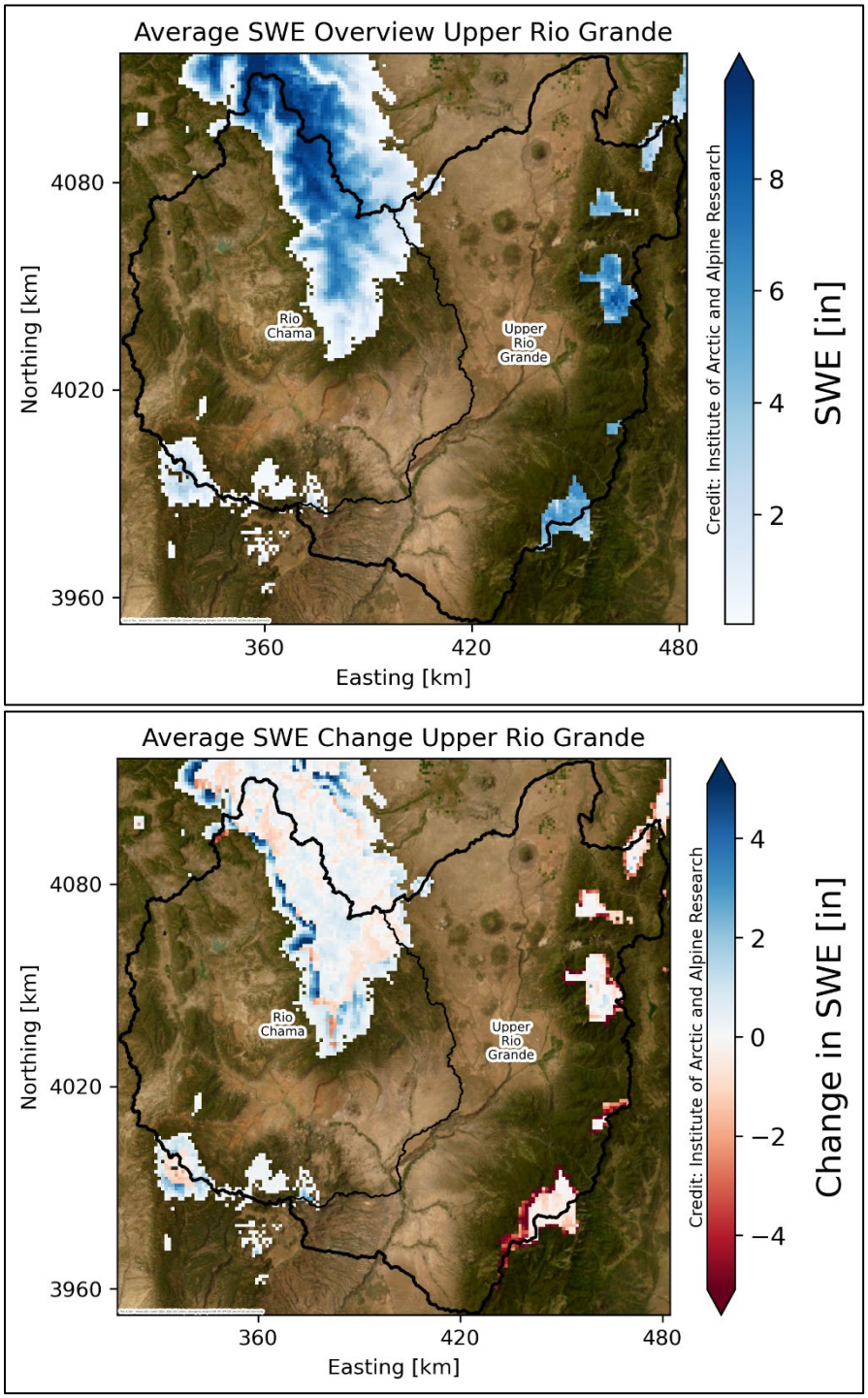

Figure 4. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Upper Rio Grande HUC6 Basin for March 08, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Upper Rio Grande Basin, showing spatial patterns of SWE across subbasins including Rio Chama and Upper Rio Grande (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Upper Rio Grande Basin from March 01, 2026, to March 08, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

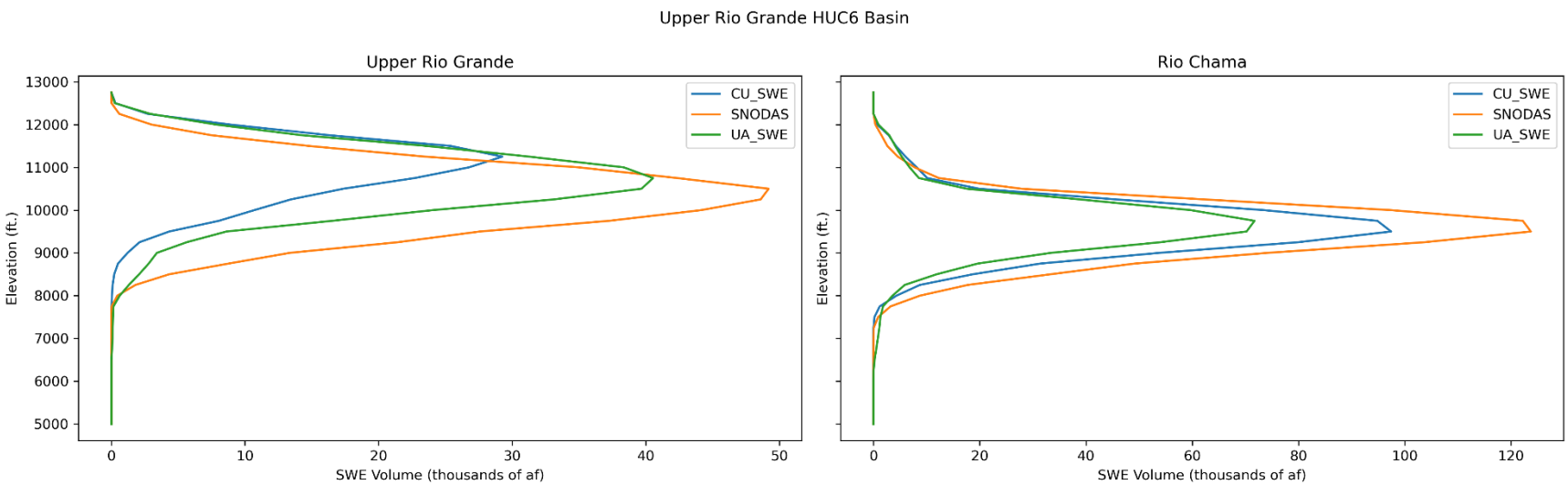

Figure 5. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Upper Rio Grande HUC6 Basin for March 08, 2026. *Elevation-distributed SWE volume for the Upper Rio Grande HUC6 Basin and associated subbasins (Rio Chama and Upper Rio Grande), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 4. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Upper Rio Grande HUC6 Basin for March 08, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since March 01, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-01 (af)	Median Vol. Change since 2026-03-01 (af)
Upper Rio Grande	Upper Rio Grande	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	31	0	0 (SNODAS, CU_SWE)	93 (UA_SWE)	31	0
		8000-9000'	369	443	42 (CU_SWE)	623 (UA_SWE)	-551	-323
		9000-10000'	5,983	3,404	1,209 (CU_SWE)	13,336 (SNODAS)	-5,529	-3,216
		10000-11000'	26,301	24,059	10,700 (CU_SWE)	44,145 (SNODAS)	-8,013	-9,757
		11000-12000'	33,364	34,976	26,768 (CU_SWE)	38,349 (UA_SWE)	750	-725
		12000-13000'	6,574	7,827	2,999 (SNODAS)	8,897 (CU_SWE)	721	1,370
	Rio Chama	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	273	9	0 (CU_SWE)	811 (UA_SWE)	273	9
		8000-9000'	5,541	4,316	3,575 (UA_SWE)	8,733 (SNODAS)	-2,418	-3,897
		9000-10000'	53,710	53,719	33,325 (UA_SWE)	74,087 (SNODAS)	-4,401	2,506
		10000-11000'	76,873	73,476	59,724 (UA_SWE)	97,420 (SNODAS)	2,715	6,457
		11000-12000'	7,551	7,731	6,789 (UA_SWE)	8,133 (CU_SWE)	319	317
		12000-13000'	650	686	342 (SNODAS)	921 (UA_SWE)	39	91

Rio Grande-Elephant Butte (Middle Rio Grande)

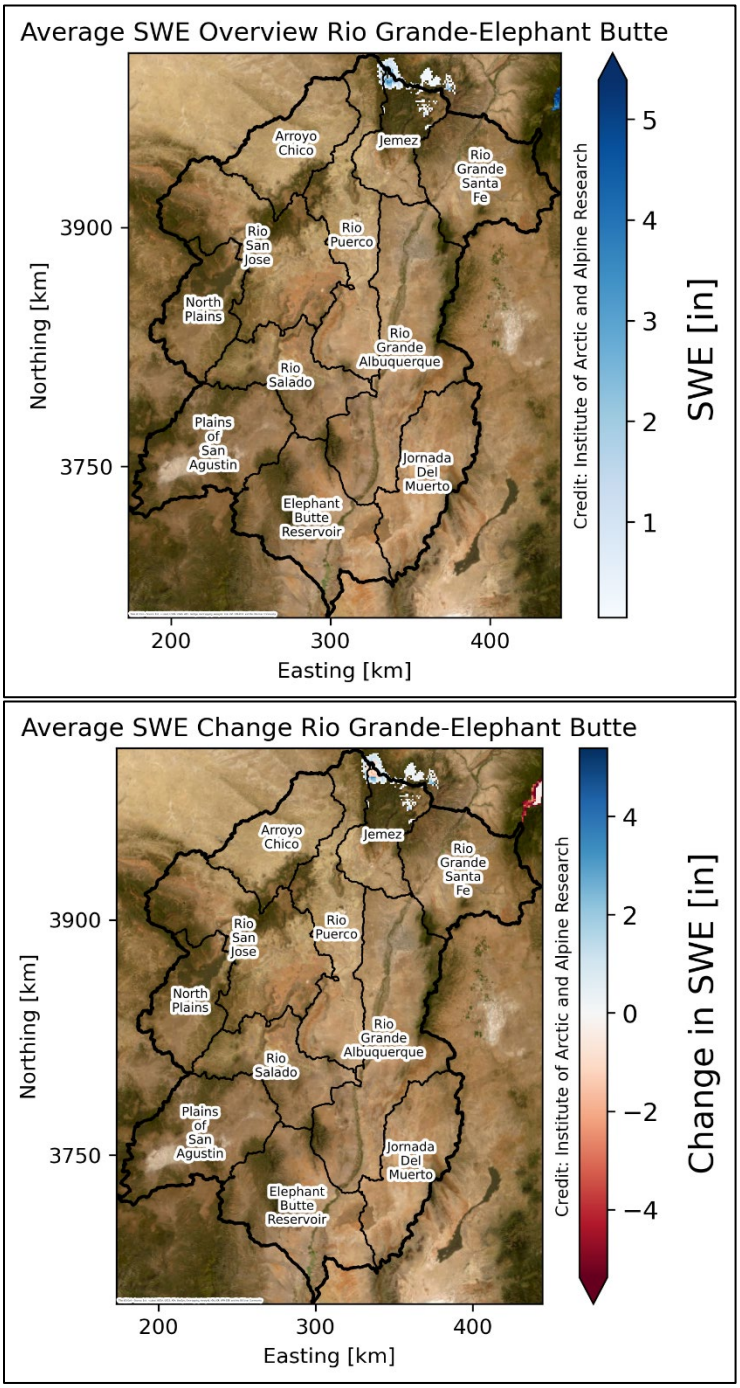

Figure 6. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande-Elephant Butte HUC6 Basin for March 08, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Rio Grande-Elephant Butte Basin, showing spatial patterns of SWE across subbasins including Rio Grande-Santa Fe, Jemez, Arroyo Chico, Rio San Jose, Rio Puerco, Rio Grande-Albuquerque, Rio Salado, North Plains, Plains of San Agustin, Elephant Butte Reservoir, and Jornada Del Muerto (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande-Elephant Butte Basin from March 01, 2026, to March 08, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

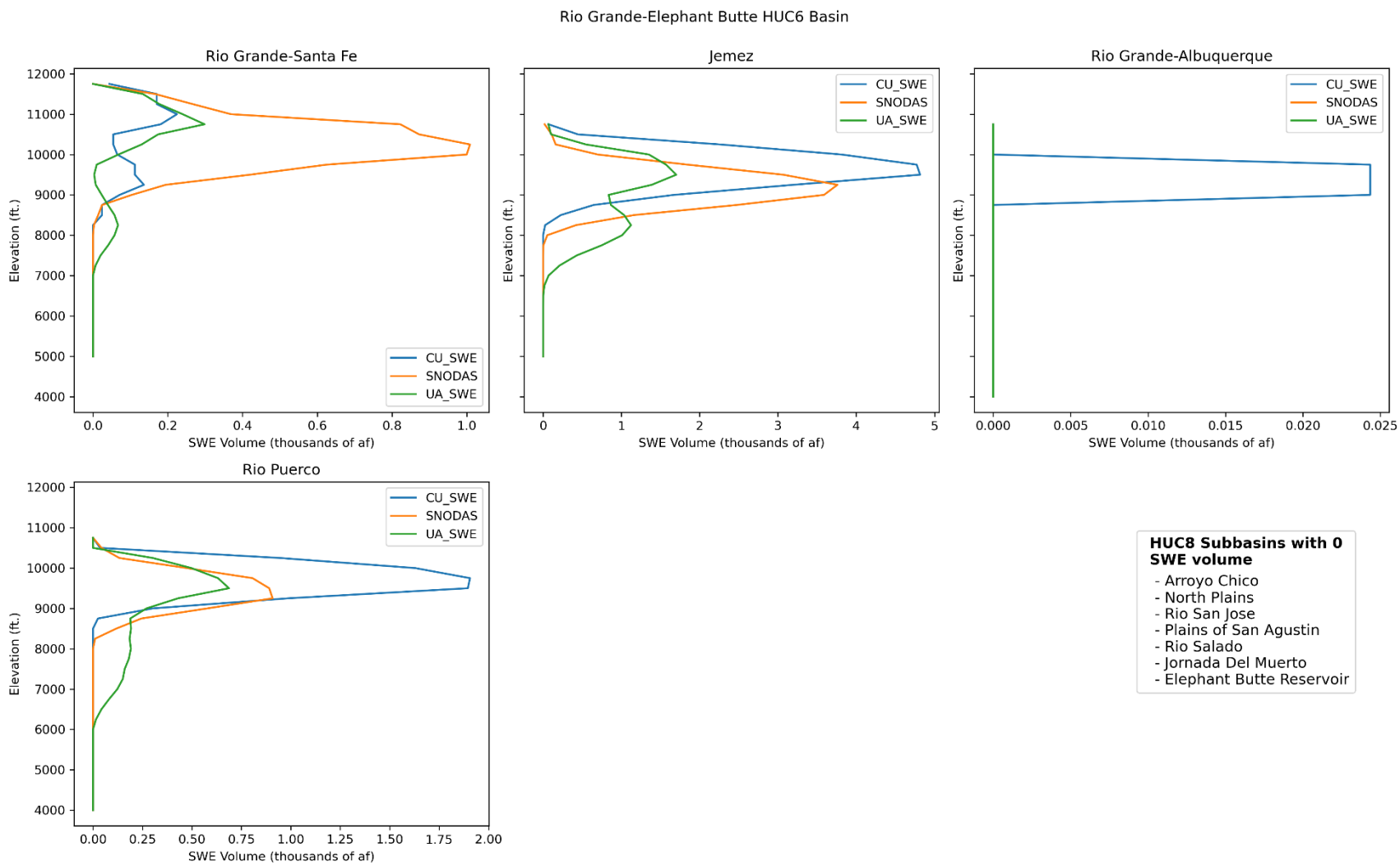

Figure 7. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Rio Grande-Elephant Butte HUC6 Basin for March 08, 2026. *Elevation-distributed SWE volume for the Rio Grande-Elephant Butte HUC6 Basin and associated subbasins (Rio Grande-Santa Fe, Jemez, Arroyo Chico, Rio San Jose, Rio Puerco, Rio Grande-Albuquerque, Rio Salado, North Plains, Plains of San Agustin, Elephant Butte Reservoir, and Jornada Del Muerto), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 5. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Rio Grande-Elephant Butte HUC6 Basin for March 08, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since March 01, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-01 (af)	Median Vol. Change since 2026-03-01 (af)
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	19	0	0 (SNODAS, CU_SWE)	58 (UA_SWE)	10	0
		9000-10000'	66	71	23 (UA_SWE)	104 (SNODAS)	-38	-57
		10000-11000'	378	68	65 (CU_SWE)	1,000 (SNODAS)	-57	-59
		11000-12000'	278	240	225 (CU_SWE)	369 (SNODAS)	-14	-49
	Jemez	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	23	0	0 (SNODAS, CU_SWE)	70 (UA_SWE)	23	0
		8000-9000'	354	54	0 (CU_SWE)	1,010 (UA_SWE)	247	-78
		9000-10000'	2,033	1,672	838 (UA_SWE)	3,589 (SNODAS)	-1,322	-2,023
		10000-11000'	1,954	1,352	698 (SNODAS)	3,814 (CU_SWE)	-321	-531
	Rio Grande-Albuquerque	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	8	0	0 (SNODAS, UA_SWE)	24 (CU_SWE)	2	0
		10000-11000'	0	0	0	0	0	0
	Arroyo Chico	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	-5	0
		10000-11000'	0	0	0	0	-17	0
	North Plains	6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-01 (af)	Median Vol. Change since 2026-03-01 (af)
Rio Grande-Elephant Butte	Rio San Jose	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	-80	0
		10000-11000'	0	0	0	0	-38	0
	Plains of San Agustin	6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	0	0
	Rio Salado	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	0	0
	Jornada Del Muerto	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
	Elephant Butte Reservoir	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	-5	0
		9000-10000'	0	0	0	0	-13	0
		10000-11000'	0	0	0	0	0	0
	Rio Puerco	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	41	0	0 (SNODAS, CU_SWE)	123 (UA_SWE)	41	0
		8000-9000'	64	0	0 (SNODAS, CU_SWE)	191 (UA_SWE)	63	0

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-01 (af)	Median Vol. Change since 2026-03-01 (af)
Rio Grande-Elephant Butte	Rio Puerco	9000-10000'	383	302	270 (UA_SWE)	578 (SNODAS)	-327	-275
		10000-11000'	867	498	473 (SNODAS)	1,629 (CU_SWE)	-212	-477

About SWEET

The following information provides a brief overview of SWEET, and the SWE products currently included in the ensemble framework. For more detailed descriptions, citations, and additional information, please refer to the GitHub repository.

SWE products currently used include:

1. **SNODAS SWE**
    a. The SNOw Data Assimilation System (SNODAS) is a physically based snow model developed by NOAA's National Operational Hydrologic Remote Sensing Center and operational since 2004. SNODAS produces daily 1-km gridded estimates of SWE and related snow variables by combining modeled weather inputs with satellite, airborne, and in situ snow observations.
2. **UA/SWANN SWE**
    a. The University of Arizona/Snow Water Artificial Neural Network Modeling System (UA/SWANN) is a research SWE product that uses snow models, artificial neural networks, and assimilated in situ measurements to generate daily gridded SWE and snow depth. The dataset provides 4-km and downscaled 800-m SWE across the conterminous United States, incorporating SNOTEL and COOP observations with PRISM precipitation and temperature fields.
3. **CU SWE**
    a. The University of Colorado (CU) SWE-Fusion product is an operational dataset produced by INSTAAR that provides near–real-time SWE estimates at 500-m resolution. The approach fuses SNOTEL observations with historical SWE reanalyses and terrain variables using a generalized linear model, and scales results using satellite-derived fractional snow-covered area from Snow Today.

The following notes describe key processing methods, assumptions, and visualization conventions used in generating SWE Evaluation Tool figures, tables, and elevation plots.

Figures:

- For ensemble figures, CU SWE and UA SWE are reprojected to the SNODAS resolution (1 sq km) and averages are only taken at pixels where at least 2 of the 3 products have SWE values greater than 0 inches.
- All overview and subbasin plots set pixels with values of 0 inches to NA so that the basemap is visible.
- For individual product figures, each product is displayed at its native resolution.
- Elevation bands of 250 feet are used in the elevation figures, and SWE volume is calculated for each product's native resolution in each band.
- Difference plots display pixels where snow was present in either the current or previous report.
- Figures are available in PNG format on the GitHub repository.

Tables:

- For all tables, mean, median, and volume are calculated at each product's native resolution - zeros are not removed from mean and median calculations.
- For ensemble tables, mean and median volume are calculated using the total volumes from the 3 products - not from an average aggregate raster.
- For min. and max., if 2 products have 0 volume, both products are listed in that cell. If all 3 products have 0 volume, no product is listed for min. or max. in that row.
- Change in SWE will only be calculated starting with the second report of the water year.
- Ensemble calculations: For HUC8 subbasins, the ensemble (mean and median) SWE and volume change (mean and median) are calculated individually for each subbasin. For the HUC6 "Total" row (tables 1 and 2), the ensemble is recalculated using the full HUC6 basin extent rather than summing HUC8. While the ensemble mean is additive and therefore consistent between HUC8 sums and the HUC6 total, the median reflects the middle value among the three products for the entire basin. As a result, median values will not sum to the HUC6 total, since they represent the difference between basin-wide medians rather than the sum of subbasin medians.
- Tables are available in CSV and PDF format on the GitHub repository.
- Elevation tables presented in the report (Tables 3, 4, and 5) are combined into one table (Table 3) on the GitHub repository.

Acknowledgments

The work completed and presented in this report is funded by NASA Western Water Actions Office (WWAO), subcontract no. 1712706, as part of the *Rio Grande River Basin Needs Assessment Report*. The SWE Evaluation Tool is a collaborative effort from members at INSTAAR, University of Colorado Boulder, and the New Mexico Office of the State Engineer.

We would also like to extend our appreciation to Noah Molotch and the Mountain Hydrology Group at CU Boulder, INSTAAR, for granting our team access to the CU SWE operational SWE dataset.